Journal of Energy Chemistry 100 (2025) 337–347

Contents lists available at ScienceDirect

Journal of Energy Chemistry

journal homepage: www.elsevier.com/locate/jechem

http://www.journals.elsevier.com/journal-of-energy-chemistry/

Machine learning empowers efficient design of ternary organic solar cells with PM6 donor

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A R T I C L E I N F O

Article history:
Received 11 June 2024
Revised 19 August 2024
Accepted 20 August 2024
Available online 7 September 2024

Keywords:
Machine learning
Ternary organic solar cells
PM6 donor
PCE

A B S T R A C T

Organic solar cells (OSCs) hold great potential as a photovoltaic technology for practical applications. However, the traditional experimental trial-and-error method for designing and engineering OSCs can be complex, expensive, and time-consuming. Machine learning (ML) techniques enable the proficient extraction of information from datasets, allowing the development of realistic models that are capable of predicting the efficacy of materials with commendable accuracy. The PM6 donor has great potential for high-performance OSCs. However, it is crucial for the rational design of a ternary blend to accurately forecast the power conversion efficiency (PCE) of ternary OSCs (TOSCs) based on a PM6 donor. Accordingly, we collected the device parameters of PM6-based TOSCs and evaluated the feature importance of their molecule descriptors to develop predictive models. In this study, we used five different ML algorithms for analysis and prediction. For the analysis, the classification and regression tree provided different rules, heuristics, and patterns from the heterogeneous dataset. The random forest algorithm outperforms other prediction ML algorithms in predicting the output performance of PM6-based TOSCs. Finally, we validated the ML outcomes by fabricating PM6-based TOSCs. Our study presents a rapid strategy for assessing a high PCE while elucidating the substantial influence of diverse descriptors.

1. Introduction

Organic solar cells (OSCs) enable low-cost electricity generation while being lightweight, semitransparent, easy to fabricate, highly flexible, and notably efficient [1–3]. In recent years, OSCs have attracted attention from the scientific community due to their diverse combination of donor (D) and acceptor (A) materials, allowing the separation of strongly bonded Frenkel excitons. Bulk heterojunction (BHJ) architectures have also been the subject of intense research because the efficiency of the charge separation process is intricately linked to the interface area between the two materials. In these architectures, the intermixed materials increase the interface area and act as transport pathways for the separated charges [4,5]. The constraints of the binary OSC can be improved by introducing a third component to the BHJ, which can function as either a secondary acceptor or donor, resulting in the creation of ternary OSCs (TOSCs). This innovative method presents possibilities for enhancing power conversion efficiency (PCE) [6]. TOSCs are mainly composed of either two donor entities and one acceptor or two acceptor entities and one donor. TOSCs provide numerous advantages, including reduced energy losses, simplified charge transport, broadened absorption range, and significantly enhanced stability, while simultaneously improving short-circuit current density (J_{SC}) and the open-circuit voltage (V_{OC}) [7]. Owing to their exceptional efficiency, ternary systems are considered to be pivotal to the future of organic photovoltaics.

V_{OC}, J_{SC}, and the fill factor (FF) are pivotal metrics that collectively determine the overall efficiency and performance of OSCs. V_{OC} represents the maximum voltage that the cell can produce when no current is flowing, directly affecting the energy output. J_{SC} is the current density generated under short-circuit conditions, reflecting the ability of the cell to capture and convert light into electrical energy. FF is an indicator of the quality of the electrical output of the cell, which is defined as the ratio of the maximum power output to the product of V_{OC} and J_{SC}, indicating how

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https://doi.org/10.1016/j.jechem.2024.08.052
2095-4956/

effectively the cell can use the generated current and voltage [8]. These parameters provide a comprehensive understanding of the factors that influence solar cell performance and efficiency, highlighting their crucial role in optimizing device functionality. Novel materials for the BHJ are typically developed using a combination of extensive trial-and-error explorations and inspiration from prior research. However, owing to the high risks of experimental failures and limited resources, conducting in-depth examinations of target materials (including the unexplored molecular space) is challenging. Considering the complexity of OSC operation, which is based on light harvesting, exciton generation, exciton diffusion, exciton separation, and carrier transport, predicting whether a material will function well based only on its chemical structure is difficult. This can be attributed to the active layer nanomorphology, whose intricate working mechanism makes design rules more challenging [9]. Accordingly, it is crucial to develop methods that shorten the OSC material development cycle to minimize the number of trial-and-error experiments and accelerate the synthesis of high-performance materials.

Machine learning (ML) has recently piqued the interest of academia and industry because of its ability to perform virtual screening at a much faster rate than quantum mechanical computations [10]. Moreover, ML algorithms can respond instantly by considering large numbers of datasets, which is not feasible for humans. Thus, ML algorithms have been employed to explore the fields of lithium-ion batteries [11,12], perovskite solar cells [13,14], light-emitting diodes [15], memory devices [16], and OSCs [17,18]. Recently, non-fullerene acceptors (NFAs) have been investigated using the ML method for the fabrication of efficient TOSCs [19]. Additionally, the V_{OC} of the TOSCs has been comprehensively investigated using double acceptors from two aspects of photovoltaic materials: doping concentration and the addition of a third component [20]. ML-assisted energy level alignment optimization is responsible for the high efficiency of non-fullerene TOSCs [21]. However, no studies have been conducted on TOSCs based on PM6, which is the most prominent donor for developing high-performance OSCs. The outstanding blend of PM6 as a donor and Y6 as an acceptor has achieved 15.7% efficiency, inspiring numerous research groups worldwide to devote their efforts to understanding and advancing these compounds [22]. PM6 [(2,6-(4,8-bis(5-(2-ethylhexyl-3-fluoro)thiophen-2-yl)-benzo[1,2-b:4,5-b0]dithiophene))-alt-(5,5-(10,30-di-2-thienyl-50,70-bis(2-ethylhexyl) benzo [10,20-c:40,50-c0]dithiophene-4,8-dione))], or PBDB-TF, or PBDB-T2F is a D-π-A copolymer with the deeper highest occupied molecular orbital (HOMO) value of −5.50 eV and an optical band gap of approximately 1.8 eV [23]. PM6 is a star polymer donor and can be blended with various NFAs, including small molecules and polymers [24,25].

In this study, we used various interpretable ML algorithms to understand the role and impact of various descriptors on TOSCs with PM6 donors. PM6 was selected due to its advantageous optoelectronic properties, including a high absorption coefficient in the visible range and well-aligned energy levels, which enhance charge transfer and separation across a wide range of acceptor materials. The compatibility of PM6 with these acceptors has resulted in the improved morphology and increased charge carrier mobility, which are both crucial for achieving high PCE [26–29]. TOSCs were selected due to their capability to extend the absorption spectrum by integrating three materials, enhancing sunlight utilization and PCE. The inclusion of a third component optimizes energy levels, improves charge transport, and reduces phase separation, resulting in lower recombination losses and superior performance. Additionally, TOSCs offer enhanced stability, longer device lifetime, and flexibility in terms of material selection, which facilitates customized performance and potentially reduces production costs [30]. These factors underscore the strategic significance of TOSCs in advancing solar cell technology. To support this concept, we collected over 14,000 data points from peer-reviewed publications and used them to analyze and predict various TOSC properties using different ML algorithms. These algorithms included classification and regression tree (CART), random forest (RF), gradient boosting (GB), linear models (LMs), and artificial neural networks (ANNs). ML-based algorithms provide an appropriate strategy for expediting the TOSC design method while assessing significant factors. The results presented herein indicate that these ML algorithms can predict the properties of TOSCs based on PM6 donors, while also providing potentially exciting possibilities to determine important descriptors for fabricating highly efficient TOSCs. To validate these methods, PM6-based TOSCs were fabricated based on the CART results, and the performance of the ML-guided TOSCs was studied. As a result, the highest PCE attained for the fabricated device was 17.21%, which corresponded to the projected value of 18%. To the best of our knowledge, this is the first report to predict the properties of TOSCs based on PM6 donors by integrating the electrical aspects of ternary blends into ML algorithms. The RF algorithm was also identified to be effective in predicting the properties of TOSCs based on PM6 donors. Ultimately, our investigation demonstrated that the RF model is effective in facilitating the design and development of highly efficient TOSCs.

2. Experimental

2.1. Establishment of database and feature extraction (descriptors)

Developing effective machine-learning models for solar cells begins with identifying key features or descriptors that influence device performance. To achieve this, we extracted crucial device parameters (including PCE, J_{SC}, V_{OC}, and FF) from the published literature, focusing on TOSCs with PM6 as a donor. These descriptors were categorized into five groups.

Solar cell structure: The key input descriptors of the device are as follows: solar cell type (conventional or inverted), anode, anode thickness, cathode, cathode thickness, electron transport layer (ETL), ETL thickness, hole transport layer (HTL), and HTL thickness. Each component is pivotal in terms of charge separation, transport, and collection. For instance, the anode and cathode materials and their thicknesses affect hole and electron collection efficiencies, while the ETL and HTL properties influence the electron and hole transport and recombination rates. Optimizing these parameters is essential to achieve a high PCE, minimal recombination losses, and optimal electrical characteristics.

Properties of ternary BHJ (t-BHJ): When discussing t-BHJ, the type of photoactive layer (PM6:A1:A2, PM6:D:A, or PM6:A: Ad (Additive)) is considered an important factor. Moreover, understanding the energy level alignment, morphology, charge transport, recombination, and light absorption is crucial. Additionally, the thickness of the photoactive or t-BHJ layer is a key consideration. The interaction between the composition and thickness of the t-BHJ significantly influences important performance metrics (such as PCE, V_{OC}, J_{SC}, and FF), ultimately resulting in enhanced solar cell performance. The components of the t-BHJ also play a vital role and should be considered. In the t-BHJ, the components with PM6 are labeled as C2 and C3.

Mixing properties: The blend or mixing proportion of the components in a t-BHJ solar cell is a crucial descriptor that significantly influences the performance of the device. This blend proportion influences energy level alignment, charge separation, morphological control, light absorption, and overall stability. Moreover, optimizing the blend proportions ensures efficient charge transfer, enhanced exciton dissociation, reduced recombination, and fine-tuning of V_{OC}, J_{SC}, FF, and PCE for optimal device performance. In this study, we extracted the data from the D: A (i.e. x:y) configuration to ensure data accuracy and completeness, because most articles reported data solely in the D: A configuration.

BHJ surface topography: Optimizing the surface topography of a t-BHJ significantly improves the TOSC performance. The surface topography influences the dissociation of excitons, pathways for charge transport, and light absorption. With optimized topography, there is improved charge collection, reduced recombination, and enhanced mechanical and thermal stability, resulting in higher efficiency and more stable solar cells.

Properties related to molecular energy levels: The descriptors used in our ML model, including HOMO and lowest unoccupied molecular orbital (LUMO) values, were extracted from a comprehensive review of previously published articles. The review encompassed both computational and experimental studies to ensure a diverse and robust dataset. LUMO level offsets (ΔLUMO) and HOMO level offsets (ΔHOMO) refer to the differences in the LUMO and HOMO energies, respectively, of the three organic molecules used in the ternary blend of a BHJ solar cell. These offsets are crucial for determining the efficiency of charge separation and transport within the solar cell. Accordingly, understanding and optimizing these offsets is essential for efficient charge transfer, reduced recombination, optimal V_{OC}, and enhanced overall performance of the solar cell. The descriptor "bandgap" refers to the energy difference between the HOMO and LUMO levels, which is also known as the HOMO-LUMO gap (ΔE_HL). This gap can be calculated by subtracting the energy of the HOMO from the LUMO (ΔE_HL = LUMO − HOMO) and represents the energy needed to excite an electron from the HOMO to the LUMO. The HOMO-LUMO gap affects the electronic properties of the molecule, such as electron affinity and ionization potential, which are essential for charge transport and separation in devices such as OSCs.

2.2. Model selection and evaluation

After analyzing the dataset using the CART algorithm, a set of rules and guidelines were obtained that provided quantitative decision rules for optimizing and designing highly efficient TOSCs for each device parameter (PCE, J_{SC}, V_{OC}, and FF). The feature importance was calculated using the RF algorithm to explore latent and possible associations between device parameters and defined descriptors, and key features were used for further analysis. In this study, ML algorithms including RF, GB, LM, and ANN were selected to build the prediction models. The following metrics were used to evaluate the models developed in this study: Pearson correlation coefficient (r), coefficient of determination (R^2), and root mean square error (RMSE). These parameters are defined as follows.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - x_i)^2}{n}} \tag{1}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(\bar{y} - x_i)^2} \tag{2}$$

$$r = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{3}$$

where x_i represents the experimental PCE extracted from the literature, $\bar{x}$ represents the average PCE, y_i represents the ML-predicted PCE, and $\bar{y}$ represents the mean value of the predicted results.

2.3. Organic solar cell fabrication

A 40-nm layer of PEDOT: PSS was deposited on a cleaned and oxygen plasma-treated ITO electrode at 3500 r min^{-1} for 40 s. Subsequently, the deposited layer underwent further annealing at 150 °C for 10 min. The photoactive layer comprising a mixture of 16 mg mL^{-1} PM6:Y6:PC$_{71}$BM (1:1.2 wt%) in chloroform and 1-chloronaphthalene (0.5%) was spin-coated at 3000 r min^{-1} for 40 s and then subsequently annealed at 90 °C for 10 min within a nitrogen-filled glove box. Prior to this step, a 5-nm layer of PNDIT-F3N was spin-coated onto the organic layer. Finally, 120-nm-thick Ag electrodes were thermally deposited using a shadow mask under a high pressure of approximately 10^{-7} Pa.

2.4. Characterization

Atomic force microscopy (AFM) in tapping mode was used to record the surface morphology of the organic layer (Park XE-100). The *I-V* test system (K3000, McScience) of the solar cell was used to determine the photovoltaic characteristics under a forward bias. The measurements were conducted under AM 1.5 (100 mW cm^{-2}) using a Keithley 2400 source measurement unit. The solar cell incident photon-to-current efficiency (IPCE) measurement system (K3100, McScience) was used to measure the external quantum efficiency (EQE) spectra of the OSCs.

3. Results and discussion

3.1. Data collection of donors and its statistics

The PM6:D:A or PM6:A:A or PM6:A:Ad (additive) dataset was manually collected (number, #462) from 236 peer-reviewed articles (Fig. S1 and Table S1). To generate the data points (approximately 14,000), 30 different variables (input: 25, output: 04, and other: 01) were used. During data collection, binary and quaternary blends based on PM6 were skimmed. To facilitate applying the ML algorithms, we encoded the extracted values into specific codes due to the complexity of using the original data. The codes for all descriptors are provided in the Supporting Data file (Excel). Specifically, we applied label encoding to incorporate categorical features into the ML models. It was evident that the number of publications and PCE based on a PM6 donor had increased over time (Fig. S1a and b), indicating the increasing interest of academia and industry in this field. Fig. S1(c) displays the distribution of the collected PCE of PM6-based TOSCs. The extensive variety of journals revealed that data points were collected from internationally reputed high-impact factor journals, implying the datasets reliability (Fig. S1d). Fig. S2 displays histograms of all the defined descriptors used for the ML algorithms. Even though J_{SC}, V_{OC}, and FF contribute mathematically equitably to the PCE (PCE = $J_{SC} \times$ FF $\times$ $V_{OC} \div$ P, where P is the amount of incident light power density), the literature suggested that J_{SC} and FF are more important than V_{OC} in the case of PM6-based TOSCs, as displayed in Fig. S3. Fig. 1 displays the typical method employed in this study. This comprised the creation of a dataset and extraction of features/descriptors, predictive ML model selection, evaluation of selected ML models, and finally, the experimental validation of the ML results.

3.2. Ml-assisted analysis and prediction of PM6-based TOSCs

We used an 80:20 split for dividing our dataset into training and testing subsets. This method ensured that 80% of the data was used for training our models across a range of ML algorithms, including CART, RF, GB, LM, and ANN. By training on the larger subset, the aim was to capture diverse patterns and relationships within the

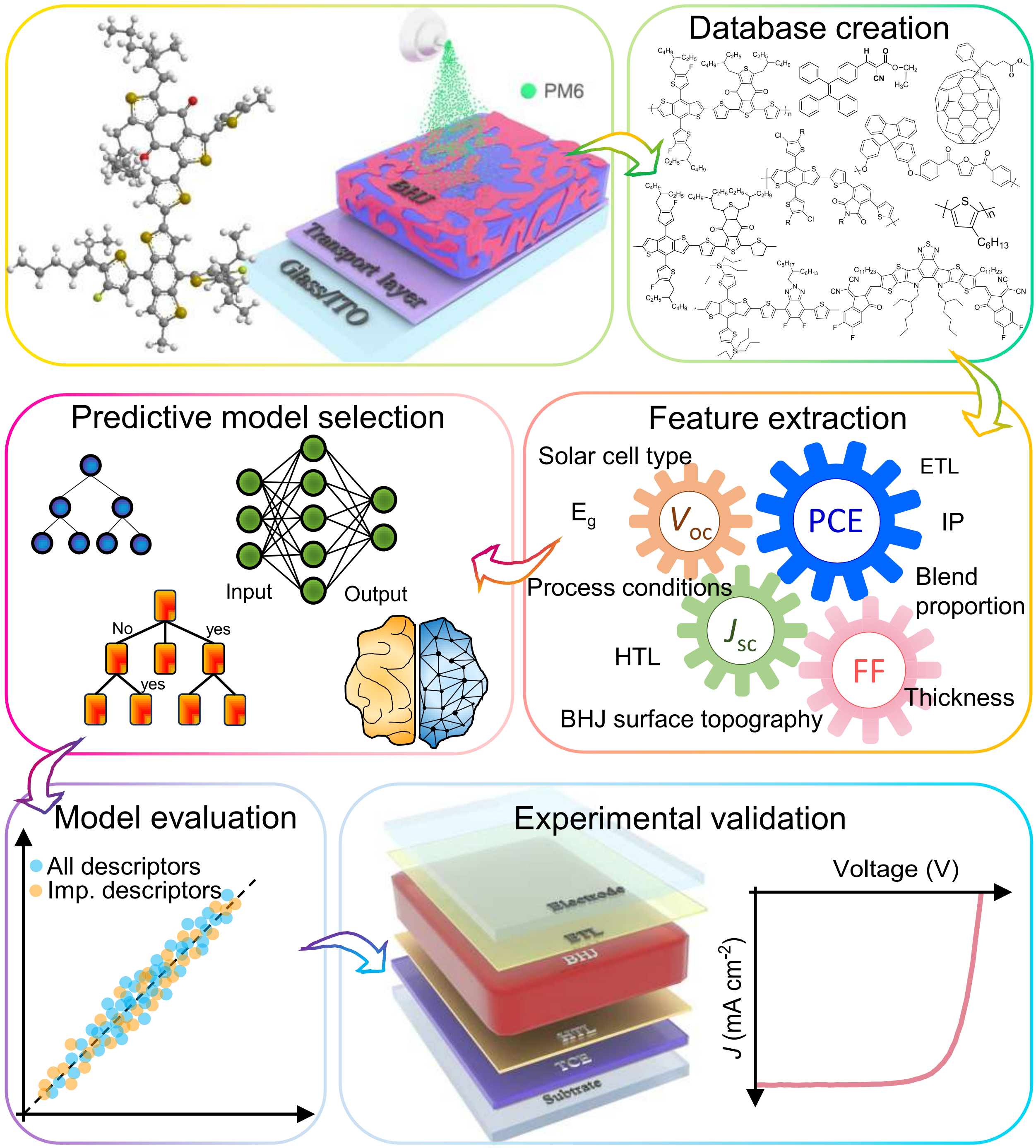

Fig. 1. Scheme of the workflow of ML-guided analysis, prediction, and fabrication of PM6-based TOSCs. Published data was used to train the model and predict device parameters for the fabrication of highly efficient OSCs.

data. The remaining 20% of the dataset was reserved for testing, enabling unbiased evaluation of each model's performance on unseen data. This approach helped to validate the robustness and generalizability of our models across different algorithmic frameworks. Initially, we used the CART algorithm for PM6-based TOSCs. CART is known as an interpretable ML algorithm because it is a decision tree algorithm that provides hidden rules, heuristics, and patterns present in the dataset [31]. Moreover, it minimizes laborious work and generates patterns from the dataset, resulting in lower computing costs and interpretable results [32]. Given these characteristics, CART was used to identify the hidden rules, heuristics, and patterns required to fabricate high-performance TOSCs based on PM6.

Fig. 2 displays the CART model of the PCE of PM6-based TOSCs. As evidenced in Fig. 2, if the blend proportion of 1:1 wt%, 1:1.5 wt%, and ≥1:1.6 wt% was satisfied, the PCE was limited to 15%. For other blend proportions, the PCE could reach 16%. Furthermore, if the device used Ag as the cathode, the PCE increased slightly, whereas the PCE decreased slightly with Al or other cathodes. According to the current CART model, a higher PCE could be obtained if the device consisted of PM6 with an electronic band gap >1.8 eV, an ETL other than metal oxides, the PFN family, or other ETLs, and a

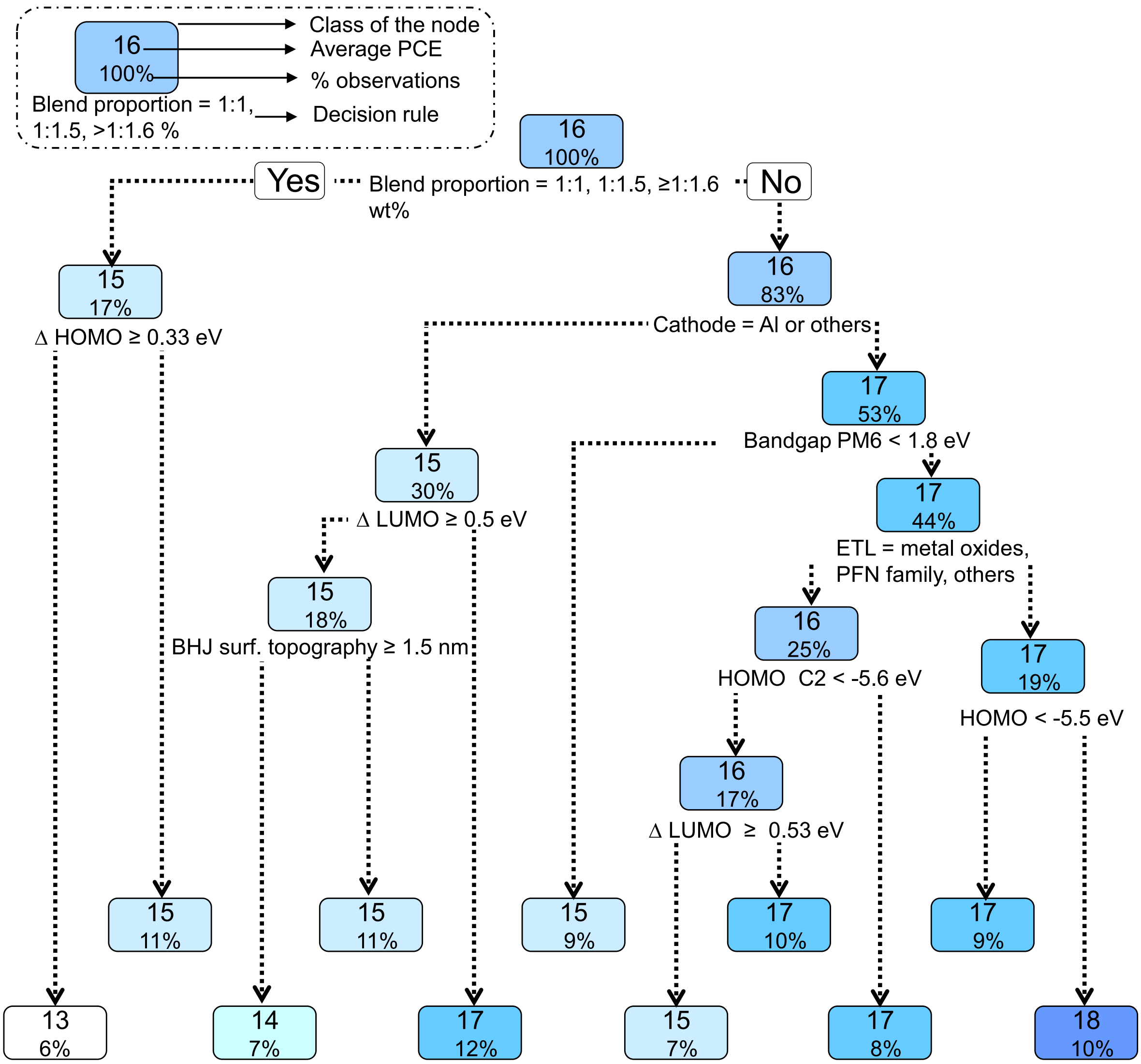

Fig. 2. CART model of PCE of the PM6-based TOSCs. For building the CART model, 25 descriptors or features were used as input variables, and PCE was considered as an output variable. The presented CART provided an average PCE value at each node, as displayed in the upper side of the box. The percentage values displayed in the boxes indicate the total number of observations following the given decision rule. The decision rule is displayed at the bottom of each node. The CART initiated with a root node and its decision rule. This was the best split for the entire dataset and provided the dominant decision rule. The root node was further divided into leaf nodes, consisting of the average PCE value and percentage observation following the given splitting and dominant decision rule.

HOMO of PM6 >−5.5 eV. The BHJ active layer was sandwiched between anode and cathode electrodes, with intermediary layers inbetween. The electrode interface significantly affects the efficiency and performance of OSCs. Therefore, establishing excellent ohmic contact is essential to ensure low resistance in the respective electrodes for optimal operation [33]. To tune the CART hyperparameters, we set the number of cross-validation folds to 10. The terminal nodes required a minimum of 25 observations, and the complexity parameter was set to 0 to control tree growth and prevent overfitting. By using information gain as the splitting criterion, the aim of the model was to enhance predictive performance and ensure robustness.

Figs. S4–S6 display the CART models of the other device parameters, such as J_{SC}, V_{OC}, and FF. As depicted in Fig. S4, if the blend proportion of the BHJ was maintained to be 1:1 wt%, the value of J_{SC} was bounded to 23 mA cm^{-2}. However, a blend proportion of 1:1.1 wt% to 1:1.7 wt% resulted in a J_{SC} value of 25 mA cm^{-2}. Additionally, if the BHJ surface topography was maintained below 2.8 nm, J_{SC} could be increased to 26 mA cm^{-2}. The orbital energies that drive further operation are the next important factor. If the LUMO C3 was higher than −4.1 eV and the HOMO C2 was higher than −5.6 eV, followed by a LUMO of PM6 higher than −3.6 eV, J_{SC} could be increased to 27 mA cm^{-2}. Finally, a material with a LUMO level below −3.9 eV was selected as C2 for the BHJ, resulting in a J_{SC} value of 27 mA cm^{-2}. From these predictions, it is evident that the BHJ composition, surface topography, and energy levels of the components in the BHJ all had a significant contribution to J_{SC}. Similarly, for V_{OC} (Fig. S5), the highest importance was assigned to blend proportion. If the blend proportion of the BHJ was 1:1.1 wt %, 1:1.2–1.29 wt%, 1:1.3 wt%, 1:1.4 wt%, or ≥1:1.6 wt% (more specifically, greater than 1:1 wt%), a V_{OC} up to 0.89 V could be achieved. It has been demonstrated that the D-A ratio significantly influences the performance of BHJ solar cells. This is because D-A ratio affects the crystalline order, phase separation, and morphology of the thin film [34]. As displayed in Fig. S6, the FF could be improved by modulating the HOMO level of C3 in the BHJ. To achieve an FF of 77%, the following conditions needed to be met. The HOMO C3 should be less than 5.6 eV, the BHJ surface roughness should be less than 1.3 nm, and the ETL should be either from PFN or PNDIT-F3N families or other electrodes. These results align with published studies suggesting that the enhanced FF is primarily attributed to the improved photo harvesting, optimized morphology for exciton dissociation, and charge transport in the TOSCs [35].

In the next stage of the study, we used different ML algorithms to predict the properties of the PM6-based TOSCs. To achieve this aim, we used four versatile ML algorithms: RF, ANN, LM, and GB. In the present case, the ANN, LM, and GB were unable to produce favorable prediction results with all the descriptors as the inputs. Table S2 presents a prediction performance comparison of the different models, with the RF method providing the best overall prediction results. The supporting material provides a detailed explanation of the ANN (Figs. S7 and S8), LM (Fig. S9), and GB (Fig. S10) ML algorithms and their results. The use of a highly heterogeneous dataset in this study could be the underlying cause of the suboptimal prediction performance of the ANN, LM, and GB models. Although heterogeneous datasets can pose challenges for ML models, they also offer several advantages, such as increased variety of information, improved generalization, enhanced feature learning, flexibility, and adaptability. Information variety can enrich the learning process by providing multiple perspectives and sources of information, potentially resulting in more comprehensive and accurate models. Moreover, ML models trained on heterogeneous datasets tend to be more flexible and adaptable to changes in the data environment. They can easily accommodate new data types or sources, allowing for seamless integration of additional information without significant retraining. ML models can leverage this diversity to learn more informative features that capture complex relationships within the data, potentially resulting in better performance compared to models trained on homogeneous datasets. Although the value of the prediction coefficient was low, it was still acceptable when compared with previous reports [36–40]. As displayed in Fig. S11, the outliers in the output parameters of the devices could potentially influence the prediction performance of the models. This is because there was a considerable difference window between the minimum values and the 1st quartile. The prediction performance could be improved by removing outliers from the dataset. Because the RF method provides good prediction results, it was selected for further feature importance analysis and prediction. Notably, the RF algorithm addresses this issue by mitigating overfitting and reducing training time, while simultaneously enhancing prediction accuracy [32,36].

Although statistical panel plots can reveal a correlation between different descriptors/features, they do not clarify how these descriptors affect device performance. To gain a better understanding of the factors that influence device performance, a feature importance score was calculated. This score can assist in determining which process conditions, compositional ratios, and device fabrication techniques have the greatest impact on device performance. Furthermore, the feature importance score can provide scientific interpretations or physical intuitions from the ML results [41]. As displayed in Fig. 3(a), the PCE of PM6-based TOSCs is highly dependent on the surface topography of the BHJ, which is determined by the role and dispersion/blending of each component that contributes to the BHJ. TOSCs can be modeled as cascade, alloy, parallel, or quadrupole, depending on the blending and energy alignments that control the mechanistic model [7]. Examining the miscibility of the components is one way of predicting the morphology type of a ternary blend [42]. Components exhibiting good miscibility typically engender an alloy-like morphology, whereas those with poor miscibility tend to adopt a parallel-like configuration [7]. Previous studies have indicated that the BHJ surface topology, blend proportion of the BHJ, and energy levels of D or A or additives have significant impacts on the PCE [19,43–45]. Fig. 3(b–d) displays a similar pattern in the other feature importance score plots, with a slight variation observed for J_{SC}, V_{OC}, and FF. Additionally, we included SHapley Additive exPlanations (SHAP) analysis plots for the PCE, J_{SC}, V_{OC}, and FF (Fig. S12) to quantify the comprehensive interpretability of the features. Although feature importance analysis identifies the most critical features, it does not indicate how these features impact device performance. SHAP analysis offers a more robust and interpretable method for understanding the impact of each feature on the model's predictions, providing a comprehensive view of feature importance. This effectively explains the ML model, even when the underlying physical principles are unknown, ensuring that the insights derived from the predictions align with actual experimental conditions. Both feature importance and SHAP analysis clearly indicated that the BHJ surface topography is the most significant factor in predicting the PCE. For J_{SC}, the blend proportion was the most influential. In the case of V_{OC}, high-impact descriptors (such as ΔLUMO and blend proportion) were key drivers. Additionally, both analyses suggested that the BHJ surface topography is also the primary driver for predicting FF. A detailed description of the SHAP explainer is provided in the Supporting Information.

Fig. 4(a) displays a correlated heat map of the PCE with important descriptors based on the feature importance results. Notably, the ETL exhibited a good correlation with the PCE, whereas other descriptors were weakly correlated. Considering these results, we used the RF algorithm to predict the PCE, *J*sc, *V*oc, and FF of PM6 donor-based TOSCs using all descriptors and important descriptors (above the group average) as inputs to the RF model. Fig. 4(b) displays a regression plot of the experimental and predicted PCEs of the PM6 donor-based TOSCs for all the defined descriptors (red spheres) and important descriptors (blue spheres) above the group average. When all the descriptors were used for prediction, the correlation between RF-predicted efficiency and experimental efficiency was moderate, with r = 0.6059. When important descriptors were used as input, the correlation coefficient was reduced to r of 0.5649. This result could have been caused by curtailed data points for prediction. Using larger and more recent datasets can improve the accuracy of ML predictions [19,45]. Researchers frequently use the RF algorithm to identify important characteristics in datasets and provide a "ground truth explanation". However, the ability of algorithm to generalize can occasionally result in outliers that significantly affect the r and R^2 scores of the model. Despite this issue, owing to its simplicity and ease of interpretation, RF remains the most suitable method for understanding the contribution of each feature to the effectiveness of a device [36]. As depicted in Fig. 4(c–e), J_{SC}, V_{OC}, and FF had comparative predictive performance (r) scores of 0.6820, 0.6068, and 0.5885, respectively, for all descriptors, and 0.6641, 0.6361, and 0.4637, respectively, for the important descriptors (Fig. 4f). This result was consistent with recent reports when all descriptors and important descriptors were considered for prediction [38,45]. The energy difference between the LUMO and HOMO levels has a noticeable impact on the ML model's performance, as it can significantly improve accuracy [45,46]. When analyzing ML models, the orbital energies of the HOMO and the LUMO must be considered. To determine the impact of energy levels on the PCE, correlation matrices are displayed in Fig. S13(a). The PCE value was highly correlated with the LUMO and HOMO levels of the C3 (r = 0.048 and 0.028) and PM6 (r = 0.034 and 0.01). This indicated that when determining the genesis of PCE in TOSCs, more attention must be devoted to the energy levels of the donor and the third component. Similar observations were made regarding J_{SC}, as displayed in Fig. S13(b). Fig. S13(c) presents the correlation matrix, which was calculated using the dataset encompassing material energy levels and V_{OC}. The LUMO value of C2 exhibited a significantly better correlation with $V_{OC,}$ while others exhibited a moderate correlation. The FF of the device was closely correlated with the HOMO level of PM6 (Fig. S13d). Fig. S14 displays schematics that clarify the

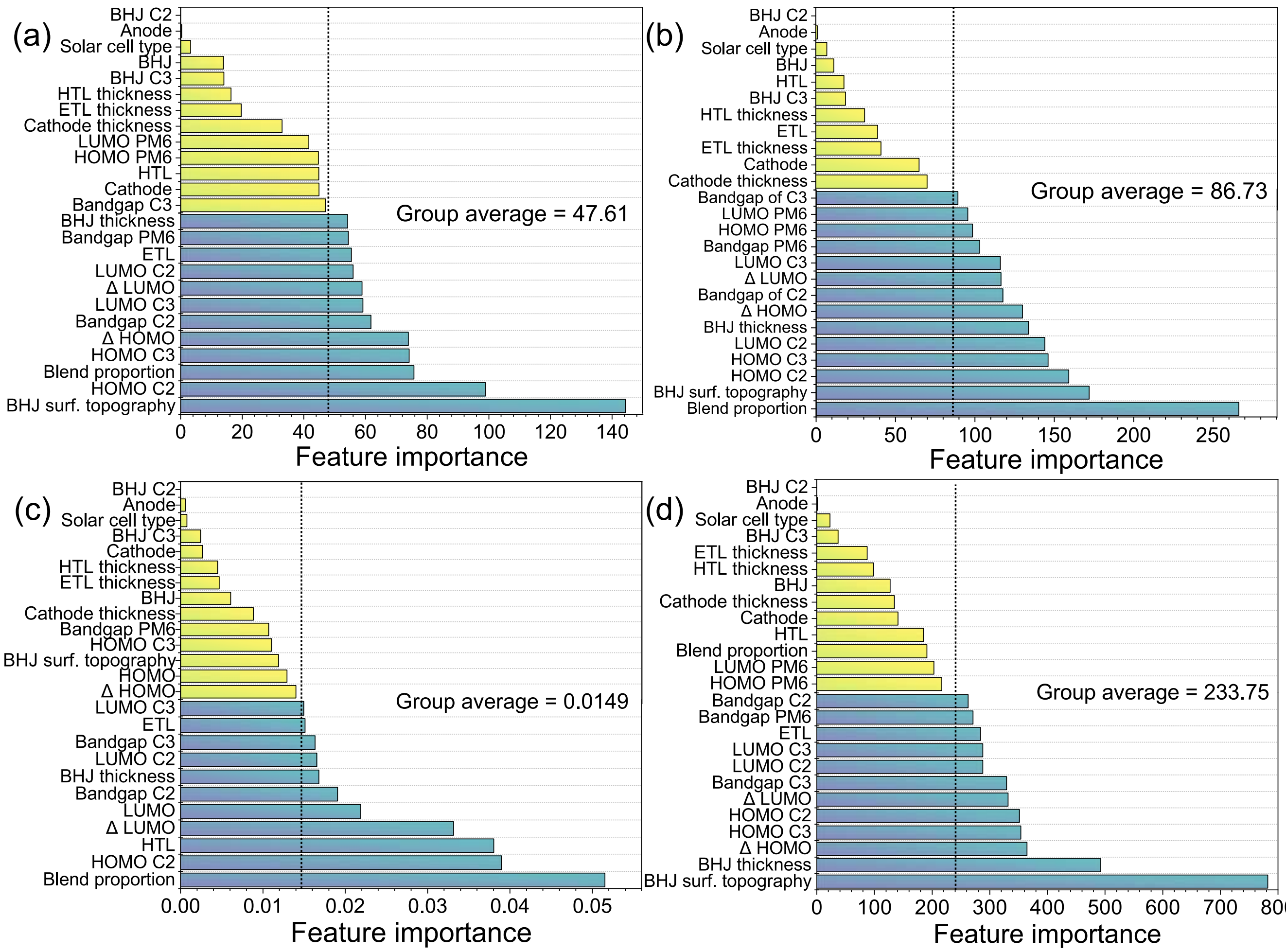

Fig. 3. Feature importance score of (a) PCE, (b) J_{SC}, (c) V_{OC}, and (d) FF. The feature importance score was calculated using the RF approach. The vertical dashed line indicates the group average. In the present case, the descriptors or features with feature importance scores greater than the group average were considered the most influential descriptors or features.

dissociation and charge transport processes in PM6/D/A and PM6/A/A configurations. Donors with finely tuned LUMO energy levels are expected to accelerate the generation of excitons at the BHJ interfaces and facilitate electron transport to the LUMO energy level.

Fig. 5 displays the joint distribution between important descriptors and device parameters, facilitating the selection of promising materials based on the molecule properties. The BHJ surface topography is considered to be the most significant and informative characteristic based on its feature importance. The PCE was revealed to be >19% when the value of surface topography was <5.93 nm, −5.8 eV < HOMO C2 < −5.26 eV, and blend proportion was ≤1:1.5 wt%, −6.21 eV < HOMO C3 < −4.88 eV. The surface topography or roughness of the best-performing t-BHJ was centered at 1.2 nm (Fig. 5a), and a blend proportion of 1:1.2–1:1.29 wt% was recommended, as depicted in Fig. 5(c).

The blend proportion is the ratio of materials used during the formation of the BHJ. A ternary blend offers a specific advantage where the third component (with high mobility) acts as a highway for charge extraction, while the other two components are primarily involved in charge splitting [47,48]. HOMO C2 (Fig. 5b) and HOMO C3 (Fig. 5d) were associated with C2 and C3, respectively, along with PM6 in the BHJ. With TOSCs, it is difficult to consider morphology, blend proportion, and charge transport separately. This is because they are intricately interconnected, influencing all stages from charge generation to charge transport and extraction. In situations where multiple molecules are combined, such as in BHJ solar cells, the energy levels are influenced not only by individual molecules but also by interactions with neighboring molecules and long-range effects arising from quadrupole moments. In contrast to a single molecule, the energy levels of solid films experience shifts because of orientation, polarization, or crystallinity effects. It was evident that the prediction of J_{SC} was primarily influenced by the proportion of the BHJ blend, the BHJ surface topography, HOMO C3, and HOMO C2 (Fig. 5e–h). As displayed in Fig. 5(i–l), for V_{OC}, in addition to the BHJ blend proportion and HOMO C2, the choice of the HTL and ΔLUMO were revealed to be important. FF prediction can be difficult because it depends on various molecular features and the morphology of the organic layer. Analysis of the empirical RF model indicated that FF was primarily determined by the BHJ surface topography, BHJ thickness, ΔHOMO, and the HOMO level of C3, as depicted in Fig. 5(m–p). All of these descriptors are intricately linked to the dimensions of an exciton/charge carrier, which includes the extraction of both holes and electrons. These factors have the potential to significantly improve the FF of TOSCs.

3.3. Experimental validation of ML predictions

To evaluate the practicability of the ML methods, we fabricated a device based on CART decision rules. This strategy allowed the PM6-based TOSCs dataset to be analyzed efficiently, providing

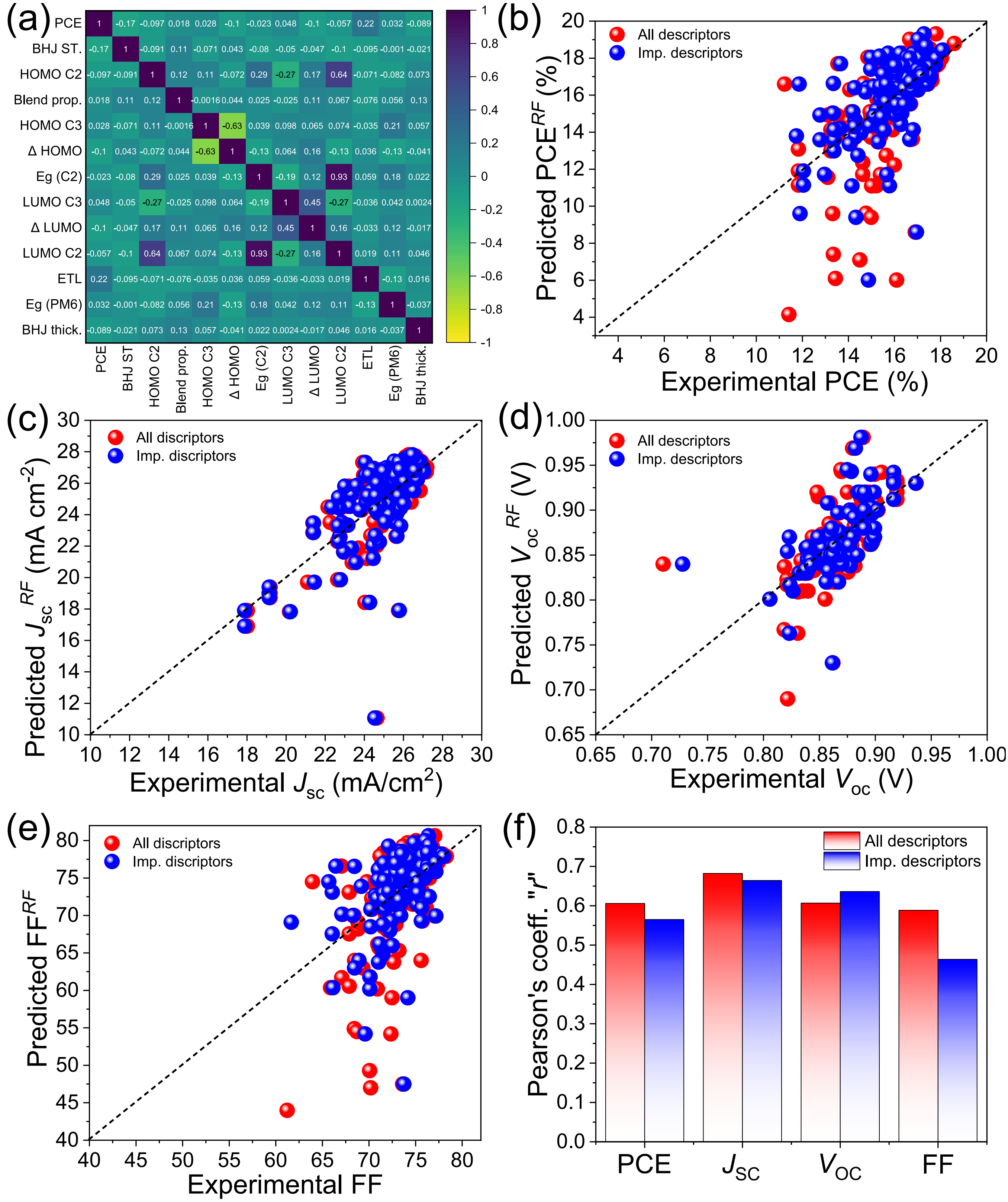

Fig. 4. (a) Pearson's correlation coefficient heat map for PCE and its important features. Predictions of (b) PCE, (c) J_{SC}, (d) V_{OC}, and (e) FF of the TOSCs based on PM6 donor using RF approach considering all defined descriptors (red spheres) and important descriptors (blue spheres) above the group average. (f) Pearson's coefficient between actual values and predictions using RF.

insights into complex ternary blend designs. The CART rule suggests using Ag as the cathode (excluding Al and other cathode materials), using the PDIN or PNDIT-F3N family as an ETL, and ensuring that the electronic bandgap of PM6 is greater than 1.8 to achieve a PCE of 18%. Considering the predicted outcomes, TOSCs with the following configurations were fabricated and mea-

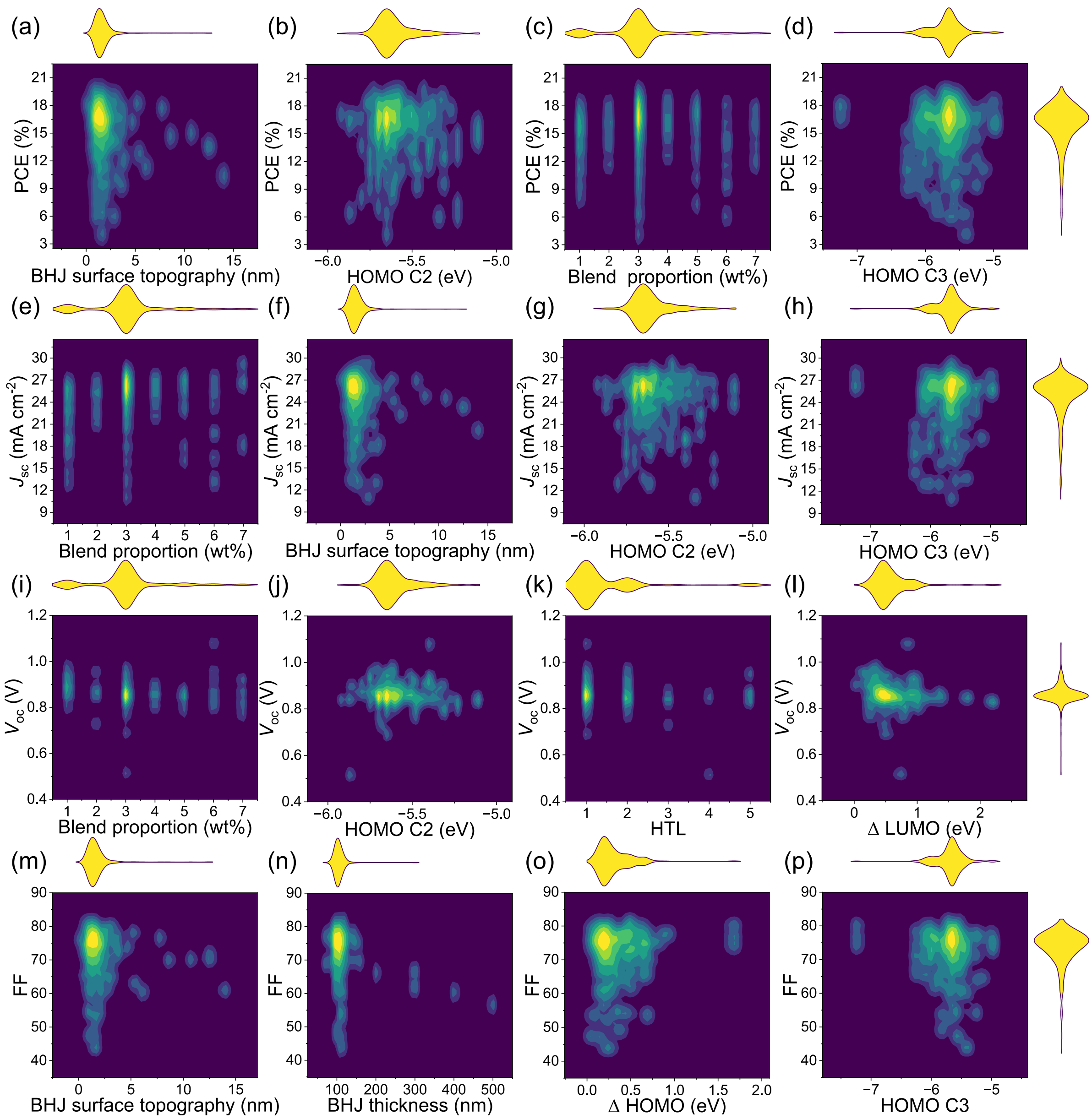

Fig. 5. Kernel density estimation based on device characteristics and vital descriptor's joint distributions. For every individual variable, there are marginal distributions. Four important descriptors determine PCE: (a) BHJ surface topography, (b) HOMO level of component 2 forming BHJ (HOMO C2), (c) blend proportion of BHJ, and (d) HOMO level of component 3 forming BHJ (HOMO C3). J_{SC} is driven by four important descriptors: (e) blend proportion of BHJ, (f) BHJ surface topography, (g) HOMO C2, and (h) HOMO C3. V_{OC} is influenced by (i) blend proportion of BHJ, (j) HOMO C2, (k) HTL, and (l) ΔLUMO. FF is mainly dependent on (m) BHJ surface topography, (n) BHJ thickness, (o) ΔHOMO, and (p) HOMO C3. Codes 1, 2, 3, 4, 5, 6 and 7 for blend proportion correspond to 1:1, 1:1.1, 1:1.2 $\leq$ BP < 1:1.3, 1:1.3, 1:1.4, 1:1.5, and $\geq$1:1.6 wt% respectively. Codes 1, 2, 3, 4, and 5 for HTL are assigned for PEDOT:PSS, MoO_3, $MO_x \neq MoO_3$, HTL free, and others, respectively.

sured: ITO/PEDOT:PSS/PM6:Y6:$PC_{71}BM$/PDIN/Ag, as displayed in Fig. 6(a). Y6 (NFA) and $PC_{71}BM$ (phenyl-C71-butyric acid methyl ester) were selected based on their structural similarity and documented performance metrics in the existing literature. Y6 is noted for its high electron mobility and strong light-absorption characteristics, rendering it particularly suitable for enhancing photocurrent generation. $PC_{71}BM$ is a widely used electron acceptor that exhibits good electron mobility and compatibility with organic donors (such as PM6). The aim of combining Y6 and $PC_{71}BM$ in ternary blends was to capitalize on their complementary absorption spectra, charge transport properties, and potential for morphology optimization to improve overall device performance. This has been substantiated by prior research in the field. As displayed in Fig. 6(b), the surface morphology of the BHJ was analyzed using AFM. The AFM image exhibited a low surface roughness with an RMS of 0.260 nm. Fig. 6(c) displays the ultraviolet-visible (UV-Vis) absorption profile for a ternary blend of PM6:Y6:$PC_{71}BM$ (Fig. S15).

Fig. 6(d) displays the proposed device's energy alignment, and Fig. 6(e) displays the electrical parameter distribution of the 10 fabricated devices. Fig. 6(f) and Table S3 present the prediction and experimental results. The majority of CART-predicted rules aligned closely with the experimentally obtained data, indicating a high level of prediction accuracy. The blend ratio was kept constant at 1:1:0.2 wt% for PM6:Y6:$PC_{71}BM$ and the device thickness was controlled up to 120 nm. Fig. S16 displays the EQE spectrum of the best-performing device, which achieved a PCE of 17.21%. The results of ML-guided PM6-based TOSCs were in good agree-

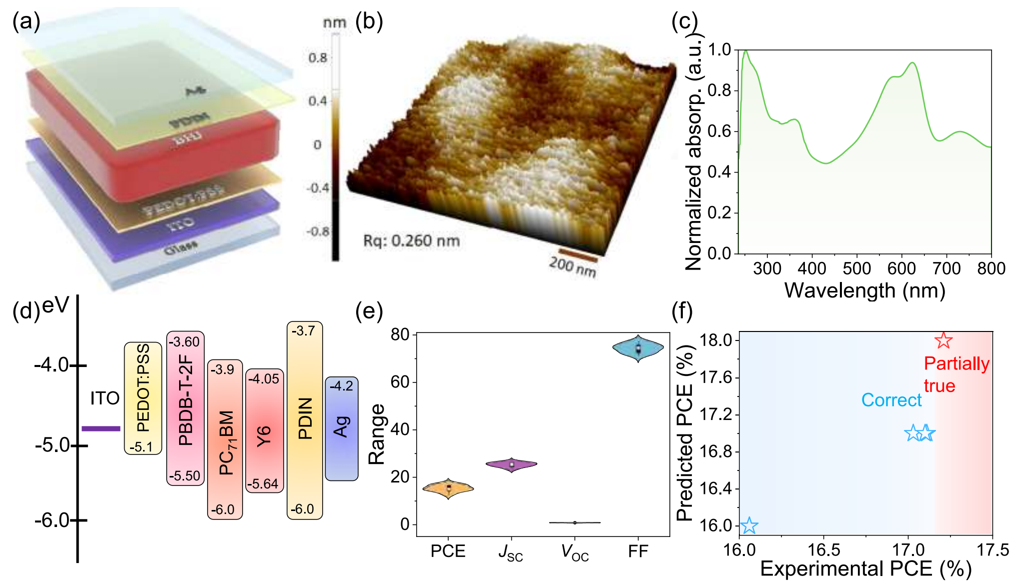

Fig. 6. (a) Device structure, (b) AFM image, (c) UV-Vis absorption spectra of the t-BHJ of the fabricated TOSC. (d) Energy band alignment, and (e) electrical parameters of the fabricated PM6 donor-based TOSCs. (f) Prediction results versus experimental data for CART-based results.

ment with the other reported devices [49–51]. According to our findings, incorporating ML algorithms for optimizing devices can reduce the need for additional experiments and speed up decision-making when fabricating highly efficient TOSCs.

4. Conclusions

Using ML methods, we were able to capture the intricate relationships between the PCE of PM6-based OSCs and important descriptors of acceptors. The CART model used approximately 14,000 data points and provided guidelines for developing highly efficient TOSCs based on a PM6 donor. The RF algorithm outperformed other predictive ML algorithms in terms of prediction accuracy. According to the statistical analysis and feature importance, the surface topography of the BHJ and the energy levels of components involved in the BHJ play a crucial role in determining the PCE values of PM6-based TOSCs. These descriptors can provide us with human-understandable insights and aid in the interpretation of the physical aspects of ML models. The results also suggested that ML algorithms can identify important parameters and provide plausible predictions for the development of novel OSC materials. The improved energy levels of the ternary active blends were primarily responsible for the performance of TOSCs from an ML perspective. These results provide a better understanding of material selection for the production of highly efficient TOSCs based on PM6.

CRediT authorship contribution statement

Kiran A. Nirmal: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tukaram D. Dongale:** Writing – review & editing, Visualization, Conceptualization. **Santosh S. Sutar:** Validation, Methodology, Investigation, Formal analysis, Data curation. **Atul C. Khot:** Visualization, Methodology, Investigation. **Tae Geun Kim:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Research Foundation of Korea (NRF) grant (No. 2016R1A3B 1908249) funded by the Korean government.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jechem.2024.08.052.

References

[1] A. Mahmood, J.L. Wang, Energy Environ. Sci. 14 (2021) 90–105.
[2] A.V. Takaloo, H.J. Lee, T.H. Park, T.D. Dongale, Y.U. Kim, D.H. Choi, T.G. Kim, J. Power Sources 515 (2021) 230589.
[3] J.W. Park, A.V. Takaloo, S.H. Kim, K.R. Son, D.Y. Kang, S.K. Kang, C.B. Lee, H. Choi, J.W. Shim, T.G. Kim, J. Power Sources 489 (2021) 229507.
[4] J.J.M. Halls, C.A. Walsh, N.C. Greenham, E.A. Marseglia, R.H. Friend, S.C. Moratti, A.B. Holmes, Nature 376 (1995) 498–500.
[5] G. Yu, J. Gao, J.C. Hummelen, F. Wudl, A.J. Heeger, Science 270 (1995) 1789–1791.
[6] N.Y. Doumon, L. Yang, F. Rosei, Nano Energy 94 (2022) 106915.
[7] M. Günther, N. Kazerouni, D. Blätte, J.D. Perea, B.C. Thompson, T. Ameri, Nat. Rev. Mater. 8 (2023) 456–471.
[8] W.B. Tarique, A. Uddin, Mater. Sci. Semicond. Process. 163 (2023) 107541.
[9] Q. Zhao, Y. Shan, C. Xiang, J. Wang, Y. Zou, G. Zhang, W. Liu, J. Energy Chem. 82 (2023) 139–147.
[10] K. Kranthiraja, A. Saeki, Adv. Funct. Mater. 31 (2021) 2011168.
[11] R.A. Eremin, P.N. Zolotarev, O.Y. Ivanshina, I.A. Bobrikov, J. Phys. Chem. C 121 (2017) 28293–28305.
[12] O. Allam, B. Woo Cho, K. Chul Kim, S. Soon Jang, RSC Adv. 8 (2018) 39414–39420.
[13] S. Lu, Q. Zhou, Y. Ouyang, Y. Guo, Q. Li, J. Wang, Nat. Commun. 9 (2018) 3405.
[14] Y. Yu, X. Tan, S. Ning, Y. Wu, ACS Energy Lett. 4 (2019) 397–404.
[15] R. Gómez-Bombarelli, J. Aguilera-Iparraguirre, T.D. Hirzel, D. Duvenaud, D. Maclaurin, M.A. Blood-Forsythe, H.S. Chae, M. Einzinger, D.-G. Ha, T. Wu, G. Markopoulos, S. Jeon, H. Kang, H. Miyazaki, M. Numata, S. Kim, W. Huang, S.I. Hong, M. Baldo, R.P. Adams, A. Aspuru-Guzik, Nat. Mater. 15 (2016) 1120–1127.
[16] T.D. Dongale, S.S. Sutar, Y.D. Dange, A.C. Khot, S.S. Kundale, S.R. Patil, S.V. Patil, A.A. Patil, S.S. Khot, P.J. Patil, J. Bae, R.K. Kamat, T.G. Kim, Appl. Mater. Today 29 (2022) 101650.
[17] E. Abbasi Jannat Abadi, H. Sahu, S.M. Javadpour, M. Goharimanesh, Mater. Today Energy 25 (2022) 100969.
[18] H. Sahu, H. Ma, J. Phys. Chem. Lett. 10 (2019) 7277–7284.
[19] M.-H. Lee, Solar RRL 7 (2023) 2300307.
[20] D. Huang, Z. Li, K. Wang, H. Zhou, X. Zhao, X. Peng, R. Zhang, J. Wu, J. Liang, L. Zhao, Polymers 15 (2023) 2954.

[21] T. Hao, S. Leng, Y. Yang, W. Zhong, M. Zhang, L. Zhu, J. Song, J. Xu, G. Zhou, Y. Zou, Y. Zhang, F. Liu, Patterns 2 (2021) 100333.
[22] J. Yuan, Y. Zhang, L. Zhou, G. Zhang, H.-L. Yip, T.-K. Lau, X. Lu, C. Zhu, H. Peng, P. A. Johnson, M. Leclerc, Y. Cao, J. Ulanski, Y. Li, Y. Zou, Joule 3 (2019) 1140–1151.
[23] M. Zhang, X. Guo, W. Ma, H. Ade, J. Hou, Adv. Mater. 27 (2015) 4655–4660.
[24] K.A. Nirmal, W. Ren, A.C. Khot, D.Y. Kang, T.D. Dongale, T.G. Kim, Adv. Sci. 10 (2023) 2300433.
[25] Q. Guo, Q. Guo, Y. Geng, A. Tang, M. Zhang, M. Du, X. Sun, E. Zhou, Mater. Chem. Front. 5 (2021) 3257–3280.
[26] Q. Fan, R. Ma, Z. Bi, X. Liao, B. Wu, S. Zhang, W. Su, J. Fang, C. Zhao, C. Yan, K. Chen, Y. Li, C. Gao, G. Li, W. Ma, Adv. Funct. Mater. 33 (2023) 2211385.
[27] B. Pang, C. Liao, X. Xu, L. Yu, R. Li, Q. Peng, Adv. Mater. 35 (2023) 2300631.
[28] T. Chen, S. Li, Y. Li, Z. Chen, H. Wu, Y. Lin, Y. Gao, M. Wang, G. Ding, J. Min, Z. Ma, H. Zhu, L. Zuo, H. Chen, Adv. Mater. 35 (2023) 2300400.
[29] C. Han, J. Wang, S. Zhang, L. Chen, F. Bi, J. Wang, C. Yang, P. Wang, Y. Li, X. Bao, Adv. Mater. 35 (2023) 2208986.
[30] Q. An, F. Zhang, J. Zhang, W. Tang, Z. Deng, B. Hu, Energy Environ. Sci. 9 (2016) 281–322.
[31] G.S. Kori, M.S. Kakkasageri, Comput. Commun. 197 (2023) 242–254.
[32] V.C. Karade, S.S. Sutar, S.W. Shin, M.P. Suryawanshi, J.S. Jang, K.S. Gour, R.K. Kamat, J.H. Yun, T.D. Dongale, J.H. Kim, Adv. Funct. Mater. 33 (2023) 2303459.
[33] W. Farooq, A.D. Khan, A.D. Khan, M. Noman, Optik 208 (2020) 164093.
[34] Y. Huang, E.J. Kramer, A.J. Heeger, G.C. Bazan, Chem. Rev. 114 (2014) 7006–7043.
[35] J. Gao, W. Gao, X. Ma, Z. Hu, C. Xu, X. Wang, Q. An, C. Yang, X. Zhang, F. Zhang, Energy Environ. Sci. 13 (2020) 958–967.
[36] M.-H. Lee, Adv. Energy Mater. 9 (2019) 1900891.
[37] H. Sahu, F. Yang, X. Ye, J. Ma, W. Fang, H. Ma, J. Mater. Chem. A 7 (2019) 17480–17488.
[38] D. Padula, J.D. Simpson, A. Troisi, Mater. Horizons 6 (2019) 343–349.
[39] B. Greenstein, G. Hutchison, J. Phys. Chem. C 13 (2023) 6179–6191.
[40] K. Wang, C. Guo, Z. Li, R. Zhang, Z. Feng, G. Fang, D. Huang, J. Liang, L. Zhao, Z. Li, Mol. Syst. Des. Eng. 8 (2023) 799–809.
[41] M.-H. Lee, Adv. Intell. Syst. 2 (2020) 1900108.
[42] P.P. Khlyabich, A.E. Rudenko, B.C. Thompson, Y.-L. Loo, Adv. Funct. Mater. 25 (2015) 5557–5563.
[43] H. Sahu, W. Rao, A. Troisi, H. Ma, Adv. Energy Mater. 8 (2018) 1801032.
[44] N. Meftahi, M. Klymenko, A.J. Christofferson, U. Bach, D.A. Winkler, S.P. Russo, npj Comput. Mater. 6 (2020) 1–8.
[45] Z.-W. Zhao, M. del Cueto, Y. Geng, A. Troisi, Chem. Mater. 32 (2020) 7777–7787.
[46] A.A. Bakulin, S.D. Dimitrov, A. Rao, P.C.Y. Chow, C.B. Nielsen, B.C. Schroeder, I. McCulloch, H.J. Bakker, J.R. Durrant, R.H. Friend, J. Phys. Chem. Lett. 4 (2013) 209–215.
[47] N. Gasparini, X. Jiao, T. Heumueller, D. Baran, G.J. Matt, S. Fladischer, E. Spiecker, H. Ade, C.J. Brabec, T. Ameri, Nat. Energy 1 (2016) 1–9.
[48] T.H. Kim, H.J. Lee, M.A. Saeed, J.H. Son, H.Y. Woo, T.G. Kim, J.W. Shim, Adv. Funct. Mater. 32 (2022) 2270174.
[49] Y. Lin, B. Adilbekova, Y. Firdaus, E. Yengel, H. Faber, M. Sajjad, X. Zheng, E. Yarali, A. Seitkhan, O.M. Bakr, A. El-Labban, U. Schwingenschlögl, V. Tung, I. McCulloch, F. Laquai, T.D. Anthopoulos, Adv. Mater. 31 (2019) 1902965.
[50] B.-H. Jiang, H.-E. Lee, J.-H. Lu, T.-H. Tsai, T.-S. Shieh, R.-J. Jeng, C.-P. Chen, ACS Appl. Mater. Interfaces 12 (2020) 39496–39504.
[51] W. Zhang, J. Huang, J. Xu, M. Han, D. Su, N. Wu, C. Zhang, A. Xu, C. Zhan, Adv. Energy Mater. 10 (2020) 2001436.